

DATABASE PROJECT REPORT: PHASE 1

Due Date: 14/09/2026 - Week 4

Project ID & Title: #6: Supply Chain & Warehouse Logistics Network Management System

A. Project Identity

- **Team Name:** G3
- **Team Members:**
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- **Project Title:** Design and Implementation of a Multi-Warehouse Logistics and Inventory Management Database System

B. Report Structure

1. Introduction & Project Scope (Adapted from ISO/IEC/IEEE 29148)

1.1 System Objective

The primary objective of the **Supply Chain & Warehouse Logistics Network Management System** is to provide a centralized, robust relational database platform to track inventory across multiple geographically distributed warehouses, manage supplier purchasing workflows, monitor multi-stage shipment logistics, and support multi-level component assembly operations.

Specifically, the system addresses key operational capabilities:

- **Multi-Warehouse Inventory Visibility:** Tracking real-time stock levels of products across different physical locations.
- **Automated Restock Threshold Tracking:** Monitoring stock levels against predefined minimum reorder thresholds to trigger inventory replenishment.
- **Bill of Materials (BOM) Management:** Supporting recursive structures where complex products are assembled from sub-components.
- **End-to-End Logistics Tracking:** Managing supplier purchase orders and tracking shipments between source and destination warehouses.

1.2 User Requirements vs. System Capabilities (ISO 29148)

A. User Requirements (UR)

- **UR-01 (Warehouse Manager):** The Warehouse Manager must be able to view inventory balances across all physical warehouses and record stock movements.

- **UR-02 (Procurement Specialist):** The Procurement Specialist must be able to create purchase orders for suppliers when product stock falls below minimum thresholds.
- **UR-03 (Logistics Coordinator):** The Logistics Coordinator must be able to dispatch shipments between warehouses and update arrival statuses.
- **UR-04 (Assembly/Production Supervisor):** The Assembly Supervisor must be able to inspect product structures to identify sub-components required for assembling higher-level finished goods.

B. System Capabilities (SC)

- **SC-01 (Data-Centric Inventory Control):** The system shall maintain an accurate quantity_on_hand for every product-warehouse pair.
- **SC-02 (Automated Restock Alerting):** The system shall evaluate stock balances against min_reorder_level and raise alerts when stock is depleted.
- **SC-03 (Recursive BOM Traversal):** The system shall support recursive relationships allowing arbitrary depths of component-subcomponent relationships.
- **SC-04 (Audit & Status Tracking):** The system shall track status updates (Pending, Completed, Canceled, In-Transit) for purchase orders and shipments.

1.3 Business Rules & Constraints (BR)

1. Inventory & Warehouse Constraints:

- **BR-101:** A warehouse must have a unique identifier (warehouse_id) and a defined storage capacity.
- **BR-102:** A product can be stored in zero or multiple warehouses. The quantity_on_hand in any warehouse cannot be negative (quantity_on_hand $\geq$ 0).
- **BR-103 (Restock Constraint):** When quantity_on_hand in a warehouse drops below the product's min_reorder_level, the system marks the item for restocking.

2. Bill of Materials (BOM) & Recursive Assembly Rules:

- **BR-201 (Recursive Structure):** A product can be composed of multiple child components, and a child component can belong to multiple parent products.
- **BR-202 (Self-Referential Integrity):** A product cannot be its own parent or child component directly or transitively (No cyclic references allowed).
- **BR-203:** The quantity_required for any component in a product structure must be strictly greater than zero (quantity_required $>$ 0).

3. Supplier Order & Purchasing Rules:

- **BR-301:** A supplier order must be associated with exactly one valid registered supplier.

- **BR-302:** An order can contain multiple line items (SupplierOrderDetail). The order_quantity must be greater than zero.
- **BR-303:** Order status must follow strict transition states: Pending → Completed or Canceled.

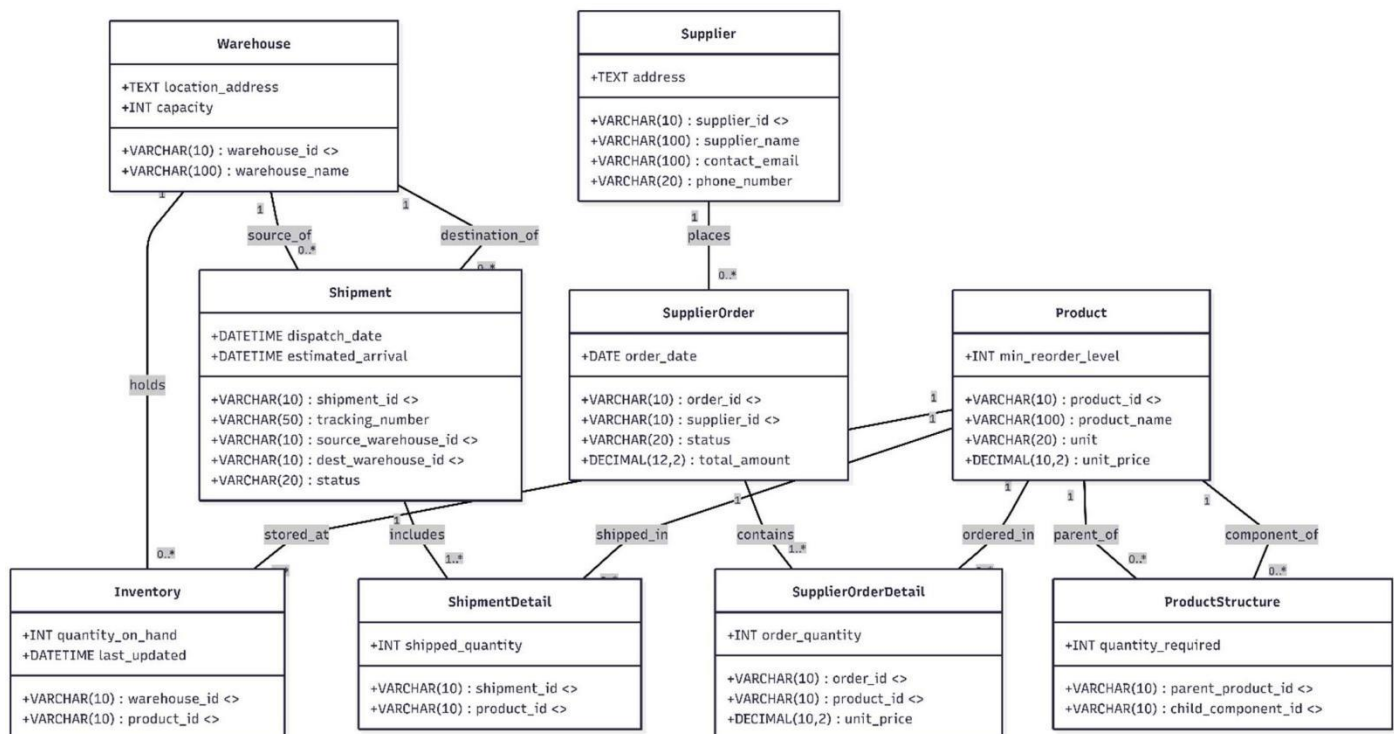
4. Shipment & Logistics Rules:

- **BR-401:** A shipment must specify a valid source warehouse (source_warehouse_id) and destination warehouse (dest_warehouse_id).
- **BR-402:** The source warehouse and destination warehouse of a shipment cannot be the same (source_warehouse_id != dest_warehouse_id).
- **BR-403:** Estimated arrival time (estimated_arrival) must be greater than or equal to the dispatch time (dispatch_date).

2. Database Design (ISO/IEC 19505 / IE Standards)

2.1 Conceptual Model (EER / UML Class Diagram)

A. Conceptual Diagram



B. Entity/Class Descriptions

1. Warehouse (Warehouse Location):

- **Description:** Represents physical warehouse facilities within the logistics network where products are stored.
- **Key attributes:** warehouse_id (PK, VARCHAR(10)), warehouse_name (VARCHAR(100)), location_address (TEXT), capacity (INT).

2. **Product (Product / Component Item):**

- **Description:** Represents inventory items, raw materials, intermediate sub-assemblies, or finished goods.
- **Key attributes:** product_id (PK,VARCHAR(10)), product_name(VARCHAR(100)), unit(VARCHAR(20)), min_reorder_level (INT), unit_price(DECIMAL(10,2)).

3. **Supplier (External Vendor):**

- **Description:** Represents external vendors providing products/materials.
- **Key attributes:** supplier_id (PK,VARCHAR(10)), supplier_name(VARCHAR(100)), contact_email(VARCHAR(100)), phone_number(VARCHAR(20)), address (TEXT).

4. **SupplierOrder (Procurement Order):**

- **Description:** Represents purchase orders issued to vendors for stock replenishment.
- **Key attributes:** order_id (PK,VARCHAR(10)), supplier_id (FK,VARCHAR(10)), order_date(DATE), status(VARCHAR(20)), total_amount(DECIMAL(12,2)).

5. **Shipment (Warehouse Inter-Transfer Logistics):**

- **Description:** Tracks product transport operations moving inventory between physical warehouses.
- **Key attributes:** shipment_id (PK,VARCHAR(10)), tracking_number(VARCHAR(50)), source_warehouse_id (FK,VARCHAR(10)), dest_warehouse_id (FK,VARCHAR(10)), dispatch_date (DATETIME), estimated_arrival (DATETIME), status (VARCHAR(20)).

6. **Inventory (Stock Balance per Location - Association Entity):**

- **Description:** Tracks real-time stock balances for specific product-warehouse pairs.
- **Key attributes:** warehouse_id (PK, FK, VARCHAR(10)), product_id (PK, FK, VARCHAR(10)), quantity_on_hand (INT), last_updated (DATETIME).

7. **ProductStructure (Bill of Materials - Recursive Association Entity):**

- **Description:** Maps parent-child assembly components for recursive Bill of Materials tracking.
- **Key attributes:** parent_product_id (PK, FK, VARCHAR(10)), child_component_id (PK, FK, VARCHAR(10)), quantity_required (INT).

8. **SupplierOrderDetail (Order Line Items - Association Entity):**

- **Description:** Represents individual item quantities and pricing within a procurement order.
- **Key attributes:** order_id (PK, FK, VARCHAR(10)), product_id (PK, FK, VARCHAR(10)), order_quantity (INT), unit_price (DECIMAL(10,2)).

9. **ShipmentDetail (Shipment Line Items - Association Entity):**

- **Description:** Details the item quantities contained inside an inter-warehouse shipment.
- **Key attributes:** shipment_id (PK, FK, VARCHAR(10)), product_id (PK, FK, VARCHAR(10)), shipped_quantity (INT).

C. Resolution of Core Database Challenges

1. Recursive Relationship (Bill of Materials - BOM Assembly):

- **Challenge:** Products can be composed of other sub-components, and sub-components can belong to multiple parent products.
- **UML Solution:** Modeled as a **Self-Association (Recursive Many-to-Many Relationship)** on the Product class via the ProductStructure association class.
- **Key Attributes:** parent_product_id (FK referencing Product), child_component_id (FK referencing Product), and quantity_required.

2. Multi-Warehouse Inventory & Restock Alerts:

- **Challenge:** Managing stock quantities across multiple warehouses and alerting when stock drops below threshold.
- **UML Solution:** Modeled via the Inventory association class linking Warehouse (1..*) and Product (0..*).
- **Key Mechanisms:** quantity_on_hand stores real-time physical balance per warehouse. Comparing quantity_on_hand (in Inventory) against min_reorder_level (in Product) provides the exact conditional trigger for automated restock alerts.

D. Multiplicity Summary

- **Supplier to SupplierOrder (1 : 0..*)**: A supplier can have zero or many purchase orders; each order belongs to exactly one supplier.
- **Warehouse to Inventory (1 : 0..*)**: A warehouse holds zero or many inventory records; each inventory record is bound to one warehouse.
- **Product to Inventory (1 : 0..*)**: A product can be stored in zero or many warehouses.
- **Warehouse to Shipment (1 : 0..*)**: A warehouse can act as the source or destination for multiple shipments.
- **Product to ProductStructure (1 : 0..*)**: A product can act as a parent to multiple sub-components and/or a sub-component to multiple parent products.